

Big Data Architecture and Governance

Individual Project | American Electric Power

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College of Engineering



1. COMPANY & SWOT ANALYSIS

Company Name: American Electric Power

Industry: Energy and Utilities

History/Background:

American Electric Power was founded in 1906 as American Gas and Electric Company. It acquired Electric Company of America in 1907, gaining control of 23 utilities across eastern and mid-west states. In 1910, the company divested gas operations to focus exclusively on electricity. Under the leadership of Philip Sporn, who became president in 1947, The company pioneered extra-high-voltage transmission technology. In 1953, it became the first utility to utilize 345 kV transmission lines, establishing technological dominance. It renamed itself AEP in 1958. From 1986-2000, the company grew by expanding its transmission network and merging with Central and South-West Corporation, expanding to 11 states. Today, AEP is a holding company organized into four business segments across its 11-state service territory, with operations in generation, transmission, and distribution. This diversified structure across 11 states reduces regulatory and geographic concentration risk.

SWOT

Strength

- Geographic diversity across 11 states
- Dominant power delivery network
- Huge customer demand
- Leading the energy transition
- Proven innovation track record

Opportunity

- Demand for data centers
- Federal government support
- Structural electricity demand growth
- New transmission line projects winning contracts

Weakness

- Aging coal infrastructure
- Regulatory penalties and uncertainties
- Customer pushback on rates increase
- Equipment sourcing challenges

Threats

- Cybersecurity attacks on power grid
- Data center customers may not materialize
- Regulatory and political uncertainty
- Revenue loss from coal plant closures



2. IMPACT STATEMENT

Impact Statement

Enterprise Data Architecture and Governance Framework

- **Project Name: Enterprise Data Architecture and Governance Foundation for Integrated Analytics**

•**Project Overview:** Designing and implementing an enterprise-scale data warehouse and governance framework that standardizes data management practices, security protocols, analytical standards across all 4 AEP business segments(Vertically Integrated Utilities, Transmission & Distribution, AEP Transmission Holdco and Generation and Marketing). Initial implementation focuses on VIU segment that has 6 operating companies, with scalable architecture supporting expansion to other segments.

•**Impact Statement:**

- **Establish Enterprise Data Foundation:** Consolidates siloed data from 6 VIU operating companies, call centers and grid operations into a unified platform. Enables consistent customer definitions, billing reconciliation and operational analytics impossible with fragmented systems..
- **Enable Data-Driven Decision Making:** Business leaders across VIU gain access to unifies, real-time insights from multiple data sources, eliminating information silos.
- **Ensure Regulatory Compliance and Security:** Standardized security protocols, access controls and data governance meet state utility commission requirements across 11 states
Clear data ownership and stewardship ensures accountability for data quality and appropriate use
- **Reduce Operational Costs :**Eliminates redundant data storage, duplicate data engineering efforts, and manual reconciliation across 6 companies.
- **Create Scalable Architecture for Future Growth:** Enterprise-designed warehouse and governance framework scales seamlessly to Transmission & Distribution, Generation & Marketing and AEP Transmission Holdco. Initial VIU implementation demonstrates viability for company-wide expansion.
- **Support Advanced Analytics:** Clean,. Governed data provides foundation for machine learning models on churn prediction, customer satisfaction and grid optimization.

•In summary, ...The Enterprise Data Warehouse transforms AEP from a federation or regional utilities into an integrated, data-driven organization. By establishing enterprise standards while implementing first in VIU, AEP creates the foundation for sophisticated analytics, improves customer experiences and informed capital investment across all business segments.....

Impact Statement

T&D Grid Optimization and Predictive Operations Platform

Project Name: Transmission and Distribution Grid Optimization and Predictive Operations Analytics Platform

Project Overview: Building a real-time analytics platform that consolidates data from SCADA systems, grid sensors, equipment monitoring, and weather data across AEP's Transmission and Distribution network to enable predictive maintenance, outage forecasting, equipment failure prediction and real time grid optimization.

Impact Statement:

- **Reduce Unplanned Outages:** Predictive equipment failure models identify impending failures SCADA data and sensor reading. Enables preventive maintenance before failure occurs. Every prevented outage protects hospital patients on life-critical equipment, preserves data center customer contracts worth hundreds of millions in annual revenue, and avoids regulatory fines across 11 states
- **Optimized Preventive Maintenance** :Analyze equipment condition, age, operating load patterns to predict remaining useful life. Optimize maintenance scheduling to prevent failures while avoiding unnecessary maintenance. Condition-based maintenance replaces calendar-based schedules — AEP stops paying to maintain equipment that does not need it and stops missing equipment that does
- **Enable Real-Time Grid Optimization:** Real-time load forecasting and system modeling optimize generation dispatch and transmission routing. Reduces transmission losses and improves system efficiency across the transmission network. Real-time dispatch recommendations reduce transmission losses and improve efficiency across the network — directly supporting the \$54B capital investment plan by making existing infrastructure perform better before new infrastructure is built.
- **Improve Grid Resilience for Data Center Growth:** The estimated 29 GW of committed data center customers require exceptional reliability. Outage prediction and prevention supports data center customer satisfaction and revenue protection. Data centers sell uptime to their own customers — hyperscalers, financial institutions, and healthcare systems. AEP's reliability directly determines whether data center customers can keep their own promises
- **Guide Capital Investment Priorities:** Distribution bottleneck analysis and load forecasting by region identify which parts of the distribution network need upgrades. Prioritizes capital investment towards highest-impact projects.
- **Support Vegetation Management:** Satellite imagery +weather +outage history identifies trees threatening power lines. Proactive trimming prevents weather-related outages
- In summary, ...The platform transforms AEP's distribution and transmission operations from reactive problem-solving to proactive, data-driven management. By predicting equipment failures, forecasting outages and optimizing grid operations in real-time, it improves reliability for the customers, reduces operational costs and optimizes capital investment for unprecedented demand growth

Why these projects?

Define the SWOT item/s related to this project and why



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- Project Name: Enterprise Data Architecture and Governance Foundation for Integrated Analytics

SWOT	Reason
Proven innovation record	It extends AEP's existing strength in technological implementation into the data domain
Fragmented data across 6 VIU operating companies creates data inconsistencies and compliance gaps	Each operating company maintains legacy systems with different data definitions and formats. This fragmentation prevents accurate reporting to 11 state utility commissions, creating audit findings and regulatory penalties
Demand for data centers	Consolidated data enables forecasting and customer analytics needed to serve unprecedented customer demand and optimize infrastructure investment
Cybersecurity attacks on power grid	Multiple legacy systems create multiple attack vectors. Inconsistent security protocols leave systems vulnerable to cyberattacks

- Project Name: Transmission and Distribution Grid Optimization and Predictive Operations Analytics Platform

SWOT	Reason
Leverages real-time SCADA and grid sensor data to enable predictive and optimization analytics	Real-time operational data provides foundation for accurate predictive models.
SCADA and sensor data quality inconsistencies; legacy transmission systems use different data formats than distribution systems complicating integrated modeling	The different data formats, difference in sensor accuracy by manufacturer reduces model reliability and requires extensive data engineering before analytics.
Demand for data centers creates a need for exceptional grid reliability and real-time optimization	Data center customer require 99.99% uptime. Grid optimization analytics that prevents outages supports customer retention and revenue protection
Sensor failures or inaccurate data compromise forecasting and optimization	ML models are only as good as input data. Any failure creates blind spots in the system. Poor data quality reduces reliability of the models leading to incorrect optimization decisions

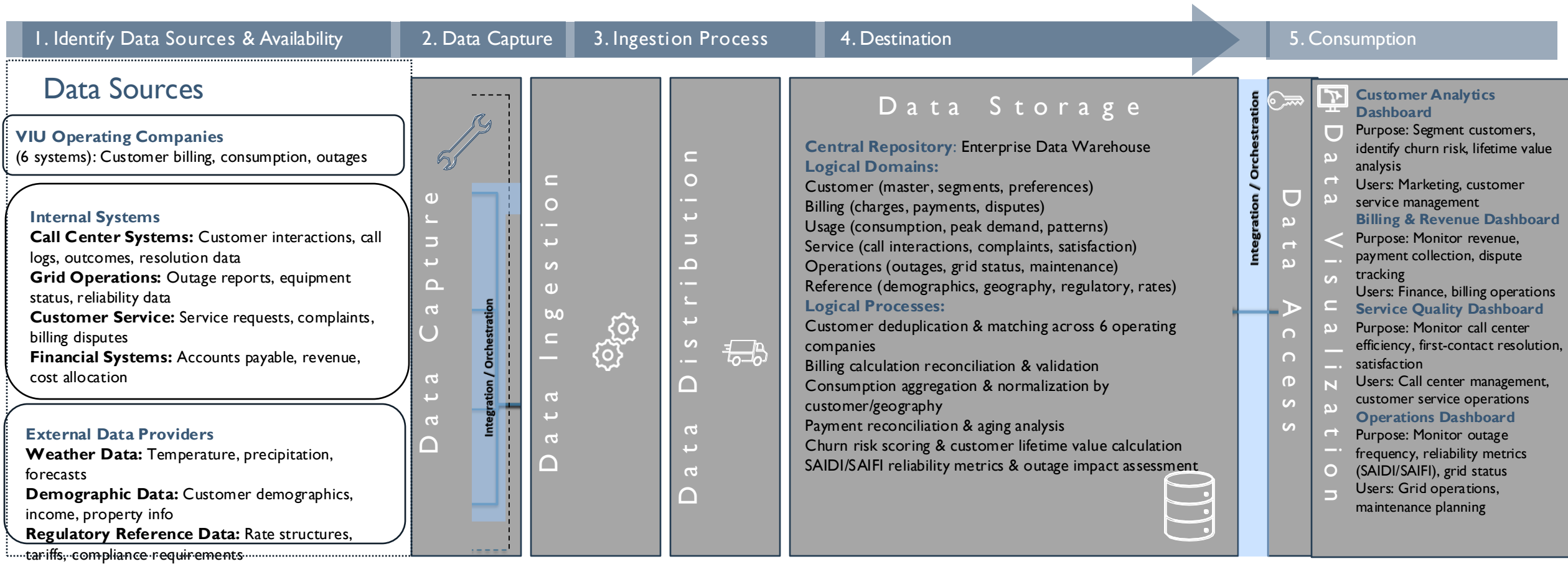


3. VISION DIAGRAM

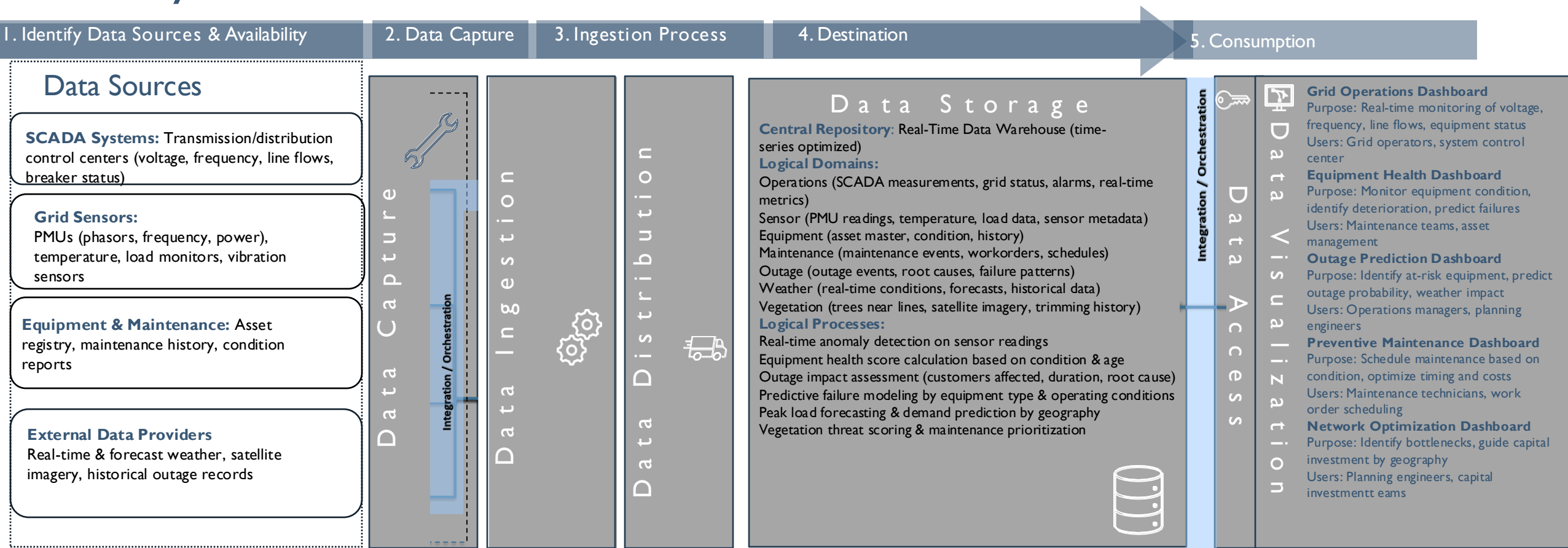
Project: Enterprise Data Architecture and Governance Foundation for Integrated Analytics



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Project: Transmission and Distribution Grid Optimization and Predictive Operations Analytics Platform





4. PROJECT DETAILS

Enterprise Data Warehouse & Governance Framework – Core Information



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- **What:** Enterprise data warehouse consolidating 6 VIU operating companies into one governed platform therefore eliminating 20+ fragmented legacy systems
- **Why now:** AEP operates across 11 states with inconsistent data standards. Fragmentation causes regulatory penalties, billing errors and inability to identify at-risk customers. The 2025 SEC settlement has placed AEP under heightened regulatory scrutiny.
- **Why \$3,560,000:** Budget built bottom-up from actual market rates. Consulting alone (\$992K) calculated as 5 specialists × 147 effective hours/month × 9 active months × \$150/hr.
- **Strategic fit:** Directly supports AEP's \$54B capital plan by establishing the data foundation needed for smarter investment decisions across all 4 business segments.
- **Timeline:** January 1 – December 31, 2026.

Project* Information

Project* Name
Enterprise Data Architecture and Governance Foundation for Integrated Analytics

Core Information **Impact** **Mandate**

Project* Required Information *

Requestor
[Dept Sponsor] Malhotra, Akash [A9937]

Detail Project* Name
Enterprise-scale data warehouse and governance framework standardizing data
Initial implementation in VIU (6 operating companies), scalable to T&D, G&M, a

Budget Information



Budget Distribution Management			
Hardware	Software	Consulting	
570,000.00	370,000.00	992,250.00	
Operations	Other	Total	
324,021.00	1,303,729.00	3,560,000.00	
Client*	GL Account	Distribution %	
American Electric Power		100.00	 
Percentage Total		100.00	

- **Hardware \$570,000** Servers, storage, backup and networking for an enterprise platform serving 3.76M customers. Redundant infrastructure required in a regulated utility environment.
- **Software \$370,000** Data warehouse platform, ETL tools, governance platform, BI dashboards and security monitoring tools across all 6 operating companies.
- **Consulting \$992,250** 5 specialists × 147 effective hours/month × 9 active months × \$150/hr. Hours adjusted for PTO. Rate reflects 2024 market rate for enterprise data integration specialists.
- **Operations \$324,021** 10% contingency buffer-industry standard for complex IT projects covering unexpected integration issues and scope changes.
- **Other \$1,303,729** People \$853K + Training \$175K + Testing \$150K + Legal \$130K (data privacy compliance across 11 states).
- **Total: \$3,560,000**

Resource Planning



g: Enterprise Data Architecture and Governance Foundation for Integrated Analytics (Start: 2026)

Category/Name	Session: 1051-60910324	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Business Analyst		1.50	1.50	1.50	1.50	1.50	1.50	1.30	1.30	0.50	0.50	0.50	0.40
Data Analyst		0.00	0.00	0.50	1.00	1.00	1.00	0.85	0.85	0.50	0.50	0.50	0.40
Data Engineer		2.00	2.00	2.00	2.50	2.50	2.50	2.10	2.10	1.00	1.00	1.00	0.80
Database Admin		1.00	1.00	1.00	1.00	1.00	1.00	0.85	0.85	1.00	1.00	1.00	0.80
Project Manager		1.00	1.00	1.00	1.00	1.00	1.00	0.85	0.85	1.00	1.00	1.00	0.80
Quality Analyst		0.50	0.50	0.50	1.00	1.00	1.00	0.85	0.85	0.50	0.50	0.50	0.00
Students		0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Systems Architect		1.00	1.00	1.00	1.00	1.00	1.00	0.85	0.85	0.50	0.50	0.50	0.40

- **Staffing Philosophy** Phased model — roles are brought in when their specific skills are needed and scaled down when their work is complete. This avoids paying full-time rates for specialists who are only needed for part of the project.
- **Phase 1 (Jan–Mar):** Design and planning. Core team active — Project Manager, Systems Architect, Data Engineers begin infrastructure setup.
- **Phase 2 (Apr–Aug):** Peak development. Full team engaged — Data Engineers scale up to 2.5, Business Analysts active, Data Governance Lead joins as framework design begins.
- **Phase 3 (Sep–Dec):** Deployment and handoff. Team scales down — most contract roles reduce as building is complete and knowledge transfer begins.
- **PTO Adjustment** July, August and December show reduced allocations (0.85/0.80) reflecting peak vacation months — not full-time assumptions.
- **Students** 0.5 FTE year-round handling data validation and documentation at significantly lower cost than senior staff.

Resource Cost Estimate

Resource Cost Estimate

				FTE Allocation												Resource Cost													
Department	Category	DailyCost	Currency	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total	
937	Business Analyst	800.0	USD	1.50	1.50	1.50	1.50	1.50	1.50	1.30	1.30	0.50	0.50	0.50	0.40	27,600.0	24,000.0	25,200.0	26,400.0	26,400.0	25,200.0	23,920.0	21,840.0	8,800.0	9,200.0	8,000.0	7,360.0	233,920.0	
937	Data Analyst	700.0	USD	0.00	0.00	0.50	1.00	1.00	1.00	0.85	0.85	0.50	0.50	0.50	0.40	0.0	0.0	7,350.0	15,400.0	15,400.0	14,700.0	13,685.0	12,495.0	7,700.0	8,050.0	7,000.0	6,440.0	108,220.0	
937	Data Engineer	600.0	USD	2.00	2.00	2.00	2.50	2.50	2.50	2.10	2.10	1.00	1.00	1.00	0.80	27,600.0	24,000.0	25,200.0	33,000.0	33,000.0	31,500.0	28,980.0	26,460.0	13,200.0	13,800.0	12,000.0	11,040.0	279,780.0	
937	Database Admin	500.0	USD	1.00	1.00	1.00	1.00	1.00	1.00	0.85	0.85	1.00	1.00	1.00	0.80	11,500.0	10,000.0	10,500.0	11,000.0	11,000.0	10,500.0	9,775.0	8,925.0	11,000.0	11,500.0	10,000.0	9,200.0	124,900.0	
937	Project Manager	800.0	USD	1.00	1.00	1.00	1.00	1.00	1.00	0.85	0.85	1.00	1.00	1.00	0.80	18,400.0	16,000.0	16,800.0	17,600.0	17,600.0	16,800.0	15,640.0	14,280.0	17,600.0	18,400.0	16,000.0	14,720.0	199,840.0	
937	Quality Analyst	500.0	USD	0.50	0.50	0.50	1.00	1.00	1.00	0.85	0.85	0.50	0.50	0.50	0.00	5,750.0	5,000.0	5,250.0	11,000.0	11,000.0	10,500.0	9,775.0	8,925.0	5,500.0	5,750.0	5,000.0	0.0	83,450.0	
937	Students	500.0	USD	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	5,750.0	5,000.0	5,250.0	5,500.0	5,500.0	5,250.0	5,750.0	5,250.0	5,500.0	5,750.0	5,000.0	5,750.0	65,250.0	
937	Systems Architect	800.0	USD	1.00	1.00	1.00	1.00	1.00	1.00	0.85	0.85	0.50	0.50	0.50	0.40	18,400.0	16,000.0	16,800.0	17,600.0	17,600.0	16,800.0	15,640.0	14,280.0	8,800.0	9,200.0	8,000.0	7,360.0	166,480.0	
																											1,261,840.0		

ROI and Cost of Ownership

On Investment Information								
Project* Estimated Cost saving information Record is Updated!								
tr	FTE	FTE Cost	Other Cost	Note	Ops Cost	Note	Updated By	Date
1	0	0.00	0.00	investment phase-no benefits realized	0.00		Kiara, Natasha	10 Mar 2026
2	0	0.00	0.00	investment phase	0.00		Kiara, Natasha	10 Mar 2026
3	0	0.00	0.00	investment phase-testing and integration	0.00		Kiara, Natasha	10 Mar 2026
4	0	0.00	0.00	investment phase-deployment prep	0.00		Kiara, Natasha	10 Mar 2026
1	1	70,000.00	1,224,625.00	early adoption-churn pred/operational efficiency	150,000.00		Kiara, Natasha	11 Mar 2026
2	2	140,000.00	1,224,625.00	growing adoption-call center effi/reg. savings	150,000.00		Kiara, Natasha	11 Mar 2026
3	3	210,000.00	1,224,625.00	full system adoption	150,000.00		Kiara, Natasha	11 Mar 2026
4	3	210,000.00	1,224,625.00	yr1 benefits realized	150,000.00		Kiara, Natasha	11 Mar 2026
1	4	260,000.00	2,222,500.00	models refined,max churn reduction,call center eff	125,000.00		Kiara, Natasha	11 Mar 2026
2	4	260,000.00	2,222,500.00	reg compliance optimized across 11 states	125,000.00		Kiara, Natasha	11 Mar 2026
3	4	260,000.00	2,222,500.00	enterprise gov framework delivering max oper value	125,000.00		Kiara, Natasha	11 Mar 2026
4	4	260,000.00	2,222,500.00	all 4 benefits are at full scale	125,000.00		Kiara, Natasha	11 Mar 2026
Total		25	1,670,000.00		13,788,500.00		1,100,000.00	

ROI and Cost of Ownership

3-Year Cost of Ownership (COO) What it costs AEP to build AND run this system over 3 years:

- 2026 Build Investment: \$3,560,000
- 2027 Operational Costs: \$600,000
- 2028 Operational Costs: \$500,000
- **Total COO: \$4,660,000**

Operational costs cover software renewals, infrastructure maintenance, support staff and security monitoring. Costs reduce in 2028 as system matures.

3-Year Return on Investment (ROI) Returns only begin in 2027 as zero benefits claimed during the 2026 build year:

- 2027 Returns: \$5,528,500
- 2028 Returns: \$9,930,000
- **Total Returns: \$15,458,500**

How 2027 returns were calculated:

- Operational efficiency: 10 staff × 50% time × \$80K × 70% adoption = \$280,000
- Churn reduction: 14,100 customers × \$85 margin = \$1,198,500
- Regulatory savings = \$300,000
- Call center efficiency: \$50M × 15% × 50% adoption = \$3,750,000

Net Value: \$10,798,500 ROI = $\$10,798,500 \div \$4,660,000 \times 100 = 232\%$ Payback: ~8 months into 2027

Impact

- **Cost of NOT Implementing: \$2,000,000** If AEP does not build this project, the annual cost of continued fragmentation includes regulatory penalties, manual reporting labor, billing errors and undetected customer churn across 6 operating companies.
- **Total Expected Saving: \$15,458,500** Combined returns over 3 years (2027 and 2028) from four benefit areas — operational efficiency, churn reduction, regulatory savings and call center efficiency. Zero benefits claimed during the 2026 build year.

Impact	Mandate		
Project* Impact			
Enhancement to the existing Bus/Ops? Yes	Current Process Improvement? Yes	Service Improvement? Yes	Cost Saving? Yes
Regulatory? Yes	Enterprise Priority? Yes	Strategic Initiative? Yes	Cost of NOT Implementing? 2000000.00
Total Expected Saving 15458500.00	Saving Years 3	Available Funding 3560000.00	

SWOT



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Strengths

1. Proven innovation track record?

Opportunities

1. Demand for data centers?

Weaknesses

1. Regulatory penalties and uncertainties ?

Threats

1. Cybersecurity attacks on power grid?

How Project I connects to AEP's strategic position:

- **S – Proven Innovation Track Record** AEP's 70-year technology leadership gives board confidence to approve \$3.6M investment and reduces political resistance from operating company staff
- **W – Regulatory Penalties Across 11 States** Unified governance eliminates inconsistent compliance reporting, transforming AEP's regulatory relationship from defensive to transparent, accelerating future capital approvals
- **O – 28 GW Data Center Demand** Consolidated customer data enables unified billing, reliability reporting and proactive service , directly protecting AEP's highest-value revenue segment
- **T – Cybersecurity Attacks** Consolidating 3.76M customer records creates a high-value target. Project replaces 20+ inconsistently secured legacy systems with enterprise-grade centralized security governance, especially critical given existing SEC scrutiny

Risk/Issues

5477	50	6	3	Integrating more than 20 legacy systems into one warehouse creates unpredictable failure points. System instability during integration could corrupt data, delay go-live, and undermine user confidence before adoption begins	01/01/2026	09/30/2026	we know failures are possible but don't know exactly what will break or when during a complex multi-system integration	Kiara, Natasha	Agreed
5476	50	9	4	Each of AEP's 6 companies has operated autonomously for years with their own data standards. Leaders may actively resist centralization to protect regional independence, causing governance deadlock and project delays	01/01/2026	03/31/2026	We know resistance will exist, but we don't know which company will resist most or if it will escalate to the leadership level. Executive sponsorship is	Kiara, Natasha	Agreed

Risk/Issues

Velero Technology Project* Portfolio Planning: [4622] Enterprise Data Architecture and Governance Foundation for Integrated Analytics

5475	75	9	6	AEP operates across 11 states, each with evolving data privacy and compliance requirements. regulatory changes mid-project may invalidate governance frameworks already built, requiring costly redesign and delaying go-live	01/01/2026	06/30/2026	known-unknown risk. we know regulations change, but we don't know which state, when, or how drastically	Kiara, Natasha	Agreed
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T&D Grid Optimization & Predictive Operations Platform – Core Information



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Project* Name: T&D Grid Optimization and Predictive Operations Analytics Platform

Planning Date: 01/01/2026

Core Information | Impact | Mandate

Project* Required Information *

Requestor: [Dept Sponsor] Malhotra, Akash [A9937]

Project Mgr*: Kiara, Natasha [A9937]

Detail Project* Name

Real-time analytics platform consolidating SCADA systems, grid sensors, equipment monitoring and weather data across AEP's T&D network to enable predictive maintenance, outage forecasting and grid optimization across 40,000+ miles of infrastructure

Project* Budget Information

Business Driver*: Strategic

Category[A]*: (none)

Category[B]*: (none)

Queue*: - Select Queue*

Documentation

Project* Web Document Location

- **What:** Real-time analytics platform connecting SCADA systems, grid sensors, equipment monitoring and weather data across AEP's 40,000+ mile T&D network
- **Why now:** AEP's grid currently operates reactively — failures are discovered after they happen. With 28 GW of data center load committing through 2030 and peak demand growing 75%, even one unplanned outage now carries major financial, regulatory and human consequences including hospitals losing power to critical care patients.
- **Why \$3,800,000:** Budget built bottom-up from actual market rates. Consulting alone (\$1.15M) reflects specialist SCADA engineers — rare professionals who understand both modern analytics and decades-old grid control systems — calculated as 4 specialists × 147 hours/month × 9 months × \$150/hr plus ML vendor engagement.
- **Strategic fit:** Directly supports AEP's \$54B capital plan — CEO Bill Fehrman stated two-thirds of the plan targets T&D infrastructure. This platform makes that investment smarter and more efficient.
- **Timeline:** January 1 – December 31, 2026.

Budget Information



Budget Distribution Management

Hardware	Software	Consulting	
750,000.00	310,000.00	1,146,600.00	
Operations	Other	Total	
345,648.00	1,247,752.00	3,800,000.00	
Client*	GL Account	Distribution %	
American Electric Power		100.00	 
Percentage Total		100.00	

- **Hardware \$750,000** Kafka streaming cluster, time-series database servers, GPU compute for ML model training and sensor gateways.
- **Software \$310,000** OSIsoft PI System (industry standard for utility operational data), Kafka enterprise support, NERC CIP security monitoring tools.
- **Consulting \$1,146,600** SCADA specialists: $4 \times 147 \text{ hrs} \times 9 \text{ months} \times \$150/\text{hr} = \$793,800$. ML vendor: $2 \times 147 \text{ hrs} \times 6 \text{ months} \times \$200/\text{hr} = \$352,800$. SCADA specialists are rare and they require both modern analytics expertise and deep knowledge of legacy grid control protocols.
- **Operations \$345,648** 10% contingency buffer covering unexpected SCADA integration complexity and scope changes.
- **Other \$1,247,752** People \$760K + Testing \$200K (includes NERC CIP compliance testing) + Training \$170K + Legal \$120K.
- **Total: \$3,800,000**

Resource Planning



Category/Name	Session: 1051-92558793	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Business Analyst		0.50	1.50	1.50	1.50	1.50	1.50	1.50	1.50	1.50	0.50	0.50	0.50
Data Analyst		0.50	0.00	0.00	0.50	1.00	1.00	1.00	1.00	1.00	0.50	0.50	0.50
Data Engineer		1.00	2.00	2.00	2.00	2.50	2.50	2.50	2.50	2.50	1.00	1.00	1.00
Database Admin		1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Project Manager		1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Quality Analyst		0.50	0.50	0.50	0.50	1.00	1.00	1.00	1.00	1.00	0.50	0.50	0.50
Students		0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Systems Architect		0.50	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.50	0.50	0.50

- **Staffing Philosophy** Phased model — specialists brought in when needed and scaled down when work is complete. Avoids paying full-time rates for roles only needed for specific phases.
- **Phase 1 (Jan–Mar): Infrastructure Setup** SCADA Engineers peak at 1.5 FTE — heaviest integration work connecting legacy grid systems.
- **Phase 2 (Apr–Aug): Model Development** Data Engineers scale up to 2.5 FTE. Data Scientist ramps in once infrastructure is ready — no point building models before data pipeline exists.
- **Phase 3 (Sep–Dec): Deployment** SCADA Engineers wind down — integration complete. Most contract roles reduce as system goes live and knowledge transfer begins.
- **PTO Adjustment** July, August and December show reduced allocations (0.85/0.80) reflecting peak vacation months.
- **Students** 0.5 FTE year-round for data labeling and QA — more available in summer so no vacation reduction applied.

Resource Cost Estimate

Resource Cost Estimate																														
Analytics Platform																														
			FTE Allocation												Resource Cost															
Category	DailyCost	Currency	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total	Base Curr		
Application Engineer	550.0	USD	1.00	1.50	1.50	1.50	1.00	1.00	0.85	0.85	0.50	0.00	0.00	0.00	12,650.0	16,500.0	17,325.0	18,150.0	12,100.0	11,550.0	10,752.5	9,817.5	6,050.0	0.0	0.0	0.0	114,895.0	114		
Business Analyst	800.0	USD	0.50	0.50	0.50	1.00	1.00	1.00	0.85	0.85	0.50	0.50	0.50	0.40	9,200.0	8,000.0	8,400.0	17,600.0	17,600.0	16,800.0	15,640.0	14,280.0	8,800.0	9,200.0	8,000.0	7,360.0	140,880.0	140		
Data Analyst	700.0	USD	0.00	0.00	0.50	1.00	1.00	1.00	0.85	0.85	0.50	0.50	0.50	0.00	0.0	0.0	7,350.0	15,400.0	15,400.0	14,700.0	13,685.0	12,495.0	7,700.0	8,050.0	7,000.0	0.0	101,780.0	101		
Data Engineer	600.0	USD	0.50	1.00	1.00	2.00	2.00	2.00	1.70	1.70	1.50	1.00	1.00	0.80	6,900.0	12,000.0	12,600.0	26,400.0	26,400.0	25,200.0	23,460.0	21,420.0	19,800.0	13,800.0	12,000.0	11,040.0	211,020.0	211		
Database Admin	500.0	USD	1.00	1.00	1.00	1.00	1.00	1.00	0.85	0.85	1.00	1.00	1.00	0.80	11,500.0	10,000.0	10,500.0	11,000.0	11,000.0	10,500.0	9,775.0	8,925.0	11,000.0	11,500.0	10,000.0	9,200.0	124,900.0	124		
Project Manager	800.0	USD	1.00	1.00	1.00	1.00	1.00	1.00	0.85	0.85	1.00	1.00	1.00	0.80	18,400.0	16,000.0	16,800.0	17,600.0	17,600.0	16,800.0	15,640.0	14,280.0	17,600.0	18,400.0	16,000.0	14,720.0	199,840.0	199		
Students	500.0	USD	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	5,750.0	5,000.0	5,250.0	5,500.0	5,500.0	5,250.0	5,750.0	5,250.0	5,500.0	5,750.0	5,000.0	5,750.0	65,250.0	65		
Test Engineer	500.0	USD	0.00	0.00	0.50	1.00	1.00	1.00	0.85	0.85	0.50	0.50	0.50	0.00	0.0	0.0	5,250.0	11,000.0	11,000.0	10,500.0	9,775.0	8,925.0	5,500.0	5,750.0	5,000.0	0.0	72,700.0	72		
																									1,031,265.0				1,031	

ROI and Cost of Ownership

Return On Investment Information

Project* Estimated Cost saving information Record is Updated!							
Year	Qtr	FTE	FTE Cost	Other Cost	Note	Ops Cost	Note
2026	1	0	0.00	0.00	Investment phase — system being built, no benefits	0.00	
2026	2	0	0.00	0.00	investment phase-pilot deployment in progress	0.00	
2026	3	0	0.00	0.00	investment phase-pilot deployment in progress	0.00	
2026	4	0	0.00	0.00	investment phase-full rollout complete	0.00	
2027	1	1	100,000.00	1,265,000.00	early adopt-ops getting confidence in predictions	150,000.00	
2027	2	2	200,000.00	1,265,000.00	growing adopt-maintenance saving/outage prevention	150,000.00	
2027	3	2	200,000.00	1,265,000.00	full adopt- all benefit areas operational	150,000.00	
2027	4	3	300,000.00	1,265,000.00	max year 1 benefits	150,000.00	
2028	1	4	400,000.00	4,630,000.00	pred models optimized, max maintenance savings	125,000.00	
2028	2	4	400,000.00	4,630,000.00	20% network coverage, 15% maintenance savings rate	125,000.00	
2028	3	4	400,000.00	4,630,000.00	max data center retention, emergency crew savings	125,000.00	

ROI and Cost of Ownership

3-Year Cost of Ownership (COO) What it costs AEP to build AND run this platform over 3 years:

- 2026 Build Investment: \$3,800,000
- 2027 Operational Costs: \$600,000
- 2028 Operational Costs: \$500,000
- **Total COO: \$4,900,000**
- Operational costs cover always-on streaming infrastructure, software renewals, support staff and mandatory NERC CIP security compliance. Costs reduce in 2028 as system stabilizes.

3-Year Return on Investment (ROI) Zero benefits claimed during the 2026 build year — returns only begin in 2027 when system is live:

- 2027 Returns: \$5,860,000
- 2028 Returns: \$20,120,000
- **Total Returns: \$25,980,000**

How 2027 returns were calculated:

- Outage prevention: 4 outages × \$200K = \$800,000
- Maintenance savings: \$500M × 10% coverage × 5% rate = \$2,500,000
- Data center retention: 8GW committed load × \$20M annual margin × 1% reliability improvement = \$560,000
- Emergency crew savings: 100 dispatches × \$20K = \$2,000,000

Why 5% savings rate in 2027? Operations teams need time to build confidence in predictive alerts before fully acting on them. Rate grows to 15% in 2028 at full adoption.

Net Value: \$21,080,000 ROI = $\$21,080,000 \div \$4,900,000 \times 100 = 430\%$ Payback: ~8 months into 2027

Impact

- **Cost of NOT Implementing: \$3,500,000** If AEP does not build this platform, the annual cost of continued reactive operations includes emergency crew overtime, unplanned equipment replacements, regulatory fines for reliability failures and data center customer attrition risk from outages. At 28 GW of committed load, even one major outage has significant financial consequences.
- **Total Expected Saving: \$25,980,000** Combined returns over 3 years (2027 and 2028) from four benefit areas — outage prevention, maintenance cost reduction, data center retention and emergency crew savings. Zero benefits claimed during the 2026 build year.

Impact	Mandate		
Project* Impact			
Enhancement to the existing Bus/Ops? Yes	Current Process Improvement? Yes	Service Improvement? Yes	Cost Saving? Yes
Regulatory? Yes	Enterprise Priority? Yes	Strategic Initiative? Yes	Cost of NOT Implementing? 3500000.00
Total Expected Saving 25980000.00	Saving Years 3	Available Funding 3800000.00	

SWOT



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Strengths

1. Dominant power delivery network?

Opportunities

1. Demand for data centers?

Weaknesses

1. Aging coal infrastructure?

Threats

1. Cybersecurity attacks on power grid?

How Project 2 connects to AEP's strategic position:

- **S** – 40,000+ miles of T&D assets generates vastly more equipment data than any comparable utility, giving predictive models more patterns to learn from, producing more accurate failure predictions across 11 states
- **W** – Aging equipment fails unpredictably today, triggering costly emergency responses and regulatory scrutiny. Project 2 detects failure signals before breakdowns occur, transforming 3am emergencies into scheduled daytime maintenance
- **O** – Data centers sell uptime to their own customers, AEP's reliability directly determines whether they can keep their promises. Project 2 makes AEP the preferred utility partner for data center expansion
- **T** – Consolidating SCADA data creates a single high-value target but enables centralized threat monitoring across all regional operations simultaneously. NERC CIP compliant security architecture built in from Day 1 — replacing inconsistently secured legacy systems

Risk/Issues

ID	Prob	Impact	Exposure	Issue/Risk	Assigned	Expected	Description*	Resource	Action
5481	50	6	3	Thousands of sensors during storm events generate unpredictable data volume spikes. If Kafka and time-series DB cannot handle peak load, prediction alerts arrive too late to be actionable. Remediation cost: \$150K–\$400 K.	01/01/2026	09/30/2026	We know infrastructure can struggle under load but don't know exactly when or how under AEP's specific real-time storm conditions.	Kiara, Natasha	Agreed
5480	50	9	4	Building accurate failure prediction models requires both ML expertise and grid domain knowledge. Models below 80% accuracy will lose operations team confidence, collapsing the ROI case before full adoption is achieved	01/01/2026	06/30/2026	We know models can underperform but don't know how they'll behave in AEP's specific grid environment until tested in production.	Kiara, Natasha	Agreed
5483	75	6	4	Each of AEP's operating companies uses different SCADA vendors, data formats, and operational standards. Conflicting standards between companies may cause data	01/01/2026	06/30/2026	We know standards will conflict but don't know whose standards will cause the	Kiara, Natasha	Agreed

Risk/Issues

				platform, undermining model accuracy and operational trust.			which company will resist standardization most.		
5479	75	9	6	AEP's legacy SCADA protocols across 7 operating companies use non-standard formats from multiple vendors and decades. Failed integration delays model development and pilot deployment, with financial exposure of \$300K–\$500K in delayed benefits.	01/01/2026	03/31/2026	we know integration will be complex but don't know exactly which systems will cause the most problems until we open them up.	Kiara, Natasha	Agreed
5482	25	100	25	Consolidating real-time SCADA data creates a high-value target. A successful attack could compromise grid controls across all 7 operating companies simultaneously, triggering outages affecting hospitals, data centers, and 5.6M customers.	01/01/2026	12/31/2026	Low probability but Critical impact makes this the highest priority risk. Consolidated platform increases the attack surface but enables better centralized defense.	Kiara, Natasha	Agreed



5. SELECTION LOGIC

Project Selection

→ Comparison Framework

	Project 1(Enterprise Framework)	Project 2(T&D Grid Optimization)
Segment	VIU (6 companies)	T&D (40,000+ miles)
Budget	\$3,560,000	\$3,800,000
Data Sources	Customer billing, consumption	SCADA, sensors, weather
Dependency	Independent	Independent

Project Selection

→ Cost & Resource Comparison

Category	Project 1(Enterprise Framework)	Project 2(T&D Grid Optimization)	Winner
Total Budget	\$3,560,000	\$3,800,000	P1
Hardware	\$570,000	\$750,000	P1
Software	\$370,000	\$310,000	P2
Consulting	\$992,250	\$1,146,600	P1
3-Year COO	\$4,660,000	\$4,900,000	P1

Project Selection

→ Timeframe & Strategic Alignment

Strategic Goal	Project 1(Enterprise Framework)	Project 2(T&D Grid Optimization)
\$54B Capital Plan	Indirect	Direct
28 GW Data Center Growth	Partial	Direct
Net-Zero 2045	Enables future	Direct foundation
NERC CIP Compliance	Partial	Direct
Regulatory Transparency	Direct	Indirect
Overall Alignment Winner		Project 2

Project Selection

Risk Assessment

Risk Type	Project 1(Enterprise Framework)	Project 2(T&D Grid Optimization)
Core Risk	Political resistance	SCADA integration
Risk Driver	Human behavior & politics	Legacy technology complexity
Controllable?	Hard to control	Solvable with specialists
Mitigation Cost	Depends on culture change	Covered by specialist contractors
DOE Grant Offset	Not applicable	\$27.8M GRIP grant available
Winner		Project 2

Project Selection

ROI & 3-Year Cost vs Benefits



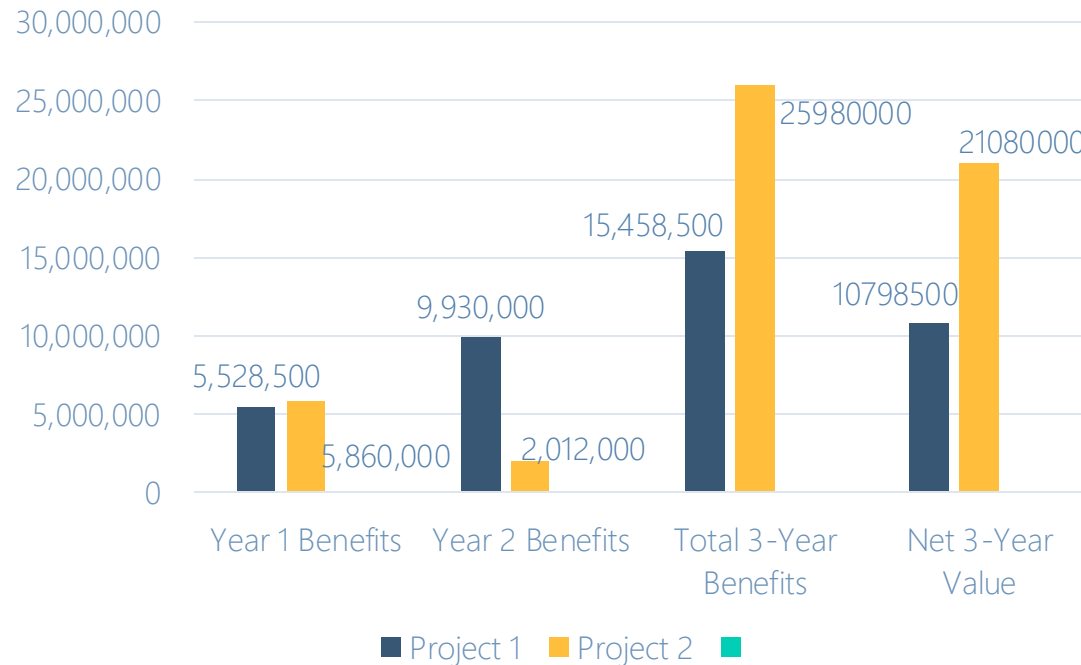
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ROI Metric	Project 1(Enterprise Framework)	Project 2(T&D Grid Optimization)	Winner
Year 1 Benefits	\$5,528,500	\$5,860,000	P2
Year 2 Benefits	\$9,930,000	\$20,120,000	P2
Total 3-Year Benefits	\$15,458,500	\$25,980,000	P2
Net 3-Year Value	\$10,798,500	\$21,080,000	P2
3-Year ROI	232%	430%	P2
Payback Period	~8 months 2027	~8 months 2027	Tie

Project Selection

ROI and COO

ROI Comparison



3-Year Cost vs Benefits



Project Selection

→ Impact Comparison

Impact Area	Project 1(Enterprise Framework)	Project 2(T&D Grid Optimization)
Customers Affected	3.76M VIU	5.6M all customers
Hospital/Critical Facilities	Indirect	Direct life-safety
Data Centers	Better billing	Guaranteed uptime
Cultural Change	Better reporting	Reactive → Proactive
Social Impact	Billing accuracy	Public safety
Winner		Project 2

Project Selection

Project Comparison Scorecard

Criteria	Weight	P1	Why	P2	Why
Cost	10%	8	Lower budget \$3.56M	6	Higher \$3.8M: SCADA specialists
Timeframe	10%	7	12 months Jan–Dec	7	Identical
Resources	10%	8	Common roles, easy to hire	6	Rare SCADA engineers
Risk	15%	6	Political - hard to control	8	Technical- solvable
Strategic Fit	20%	6	Internal governance only	9	Directly supports \$54B plan
ROI	20%	6	232% - strong	9	430% - nearly double
Impact	15%	6	3.76M customers, internal	9	5.6M customers, public safety
TOTAL	100%	47/70		54/70	Project 2 wins

Selection Rationale

Selected Project: T&D Grid Optimization & Predictive Operations Platform

Why Project 2 is the best choice:

Independence: Project 2 draws from SCADA systems, grid sensors and weather data, entirely different from Project 1's customer billing and consumption data. There is no foundation dependency and no sequencing risk. Both projects can continue independently

Urgency: AEP's most pressing challenge is managing 28 GW of committed data center load through 2030 alongside 75% demand growth. This demands grid reliability, which Project 2 delivers directly and at once

Superior ROI : 430% vs 232% over 3 years. Project 2 generates \$10.3M more in net value for only \$240K more in investment. The premium pays for itself many times over.

Broader Impact: Project 2 protects all 5.6M customers, secures hospital patient safety, fulfills data center uptime commitments, and shifts AEP from reactive to proactive operations, building the public trust that underpins AEP's social license to operate

Conclusion and Next Steps

After evaluating both projects across 7 criteria using quantitative data and business logic, T&D Grid Optimization & Predictive Operations Platform is the best choice. It delivers better ROI (430% vs 232%), broader customer impact (5.6M vs 3.76M customers), stronger strategic alignment with AEP's \$54B capital plan, and more controllable risks. The \$240K cost premium is fully justified by \$10.3M in added net value over 3 years.

Project I is not abandoned; its governance and data foundation work stays valuable for AEP's VIU segment and will be pursued in a later phase.

Next Steps for Implementation:

- Initiate procurement process for SCADA integration specialists and ML analytics vendor
- Establish project governance structure with T&D leadership sponsorship
- Conduct SCADA system inventory audit across all regions to confirm integration complexity and refine budget assumptions



FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

Project Functional Requirements

ID	Description
FR 01	The system must ingest real-time sensor data from SCADA systems across all AEP operating company control centers using standard utility protocols (DNP3, Modbus, IEC 61850)
FR 02	The system must process and store readings from PMU devices, temperature monitors, load sensors and vibration sensors across the T&D network
FR 03	The system must integrate historical maintenance records, equipment inspection reports and work order data from AEP's existing asset management systems
FR 04	The system must ingest real-time and forecast weather data from third-party providers for all regions within AEP's 11-state service territory
FR 05	The system must integrate satellite imagery data to support vegetation management analysis along transmission and distribution lines
FR 06	The system must apply data quality validation rules: flagging sensor readings outside physically possible ranges, missing data gaps and temporal inconsistencies, before any data is processed or stored
FR 07	The system must calculate equipment health scores (0–100) for each monitored asset using a weighted algorithm combining sensor readings (40%), equipment age (20%), operating load patterns (25%) and historical failure frequency (15%)
FR 08	The system must classify equipment failure risk into four categories: Critical (within 24 hours), High (within 7 days), Medium (within 30 days) and Low (beyond 30 days) — based on predictive model outputs

Project Functional Requirements

ID	Description
FR 09	The system must generate failure prediction reports showing predicted time to failure for at-risk equipment with confidence intervals
FR 10	The system must recalculate outage risk scores every 6 hours under normal conditions and every 30 minutes during active storm events
FR 11	The system must apply seasonal adjustment factors to failure predictions — accounting for higher failure rates during extreme heat and cold periods based on historical data
FR 12	The system must calculate maintenance cost savings by comparing actual condition-based maintenance costs against estimated reactive emergency repair costs for the same equipment
FR 13	The system must calculate and track grid reliability metrics including SAIDI (System Average Interruption Duration Index) and SAIFI (System Average Interruption Frequency Index) by region
FR 14	The system must generate vegetation threat scores for monitored line segments ranking areas by risk of tree-related outages based on satellite imagery, weather data and historical outage records
FR 15	The system must retrain predictive models automatically on a monthly basis using new sensor readings and maintenance outcome data to maintain accuracy over time
FR 16	The system must generate real-time alerts when equipment sensor readings exceed defined thresholds, notifying relevant users via dashboard popup, email and SMS within 90 seconds of anomaly detection

Project Functional Requirements

ID	Description
FR 17	The system must display alerts with equipment ID, location, severity level, predicted impact and recommended action — providing sufficient information for immediate decision-making
FR 18	The system must automatically escalate Critical risk equipment alerts to the on-call grid operations manager regardless of time of day
FR 19	The system must re-notify operators if a critical alert is not acknowledged within 10 minutes escalating to the next available operator after 20 minutes
FR 20	The system must generate storm preparation alerts identifying highest-risk equipment before forecasted severe weather events
FR 21	The system must automatically generate condition-based maintenance work orders when equipment health scores fall below defined thresholds
FR 22	The system must provide a real-time operational dashboard displaying grid status, active alerts, equipment health and outage risk across AEP's full T&D network on a single unified interface
FR 23	The system must display equipment health scores, historical sensor trend charts and predicted failure timelines for any individual asset within 3 clicks from the main dashboard
FR 24	The system must display a ranked list of highest outage risk equipment updated daily showing probability score, predicted failure window and customer impact

Project Functional Requirements

ID	Description
FR 25	The system must provide maintenance scheduling views showing planned, in-progress and completed activities with failure diagnosis, root cause analysis and recommended repair actions
FR 26	The system must display network bottleneck analysis and geographic heat maps identifying transmission and distribution segments operating above safe capacity thresholds
FR 27	The system must provide capital investment prioritization reports ranking infrastructure upgrade needs by risk score, customer impact and cost-benefit analysis
FR 28	The system must generate automated daily operational reports summarizing alerts triggered, maintenance actions taken and outages prevented — delivered by 6am each morning
FR 29	The system must produce monthly reliability performance reports showing SAIDI and SAIFI metrics by region for submission to state utility commissions across 11 states
FR 30	The system must generate quarterly ROI reports showing actual benefits realized — outages prevented, maintenance savings and emergency crew reductions — against projected figures
FR 31	The system must produce NERC CIP compliance reports on demand showing security controls, audit trail summaries and any compliance gaps
FR 32	The system must provide an executive summary view showing outages prevented, maintenance savings, customer reliability metrics and ROI in non-technical language for C-suite users

Project Functional Requirements

ID	Description
FR 33	The system must allow users to create custom reports by selecting data fields, time periods and geographic filters without requiring technical assistance
FR 34	The system must require multi-factor authentication for all user logins — username/password plus a second verification factor
FR 35	The system must enforce role-based access control with five defined roles: Grid Operator, Maintenance Technician, Planning Engineer, System Administrator and Executive
FR 36	The system must automatically log out inactive sessions after 15 minutes for grid operators and 30 minutes for all other roles
FR 37	The system must require supervisor approval before any system-generated maintenance work order is dispatched to field crews
FR 38	The system must support a four-stage maintenance workflow: Predicted → Scheduled → In Progress → Completed with mandatory status updates at each stage
FR 39	The system must prevent field technicians from closing work orders without entering actual repair outcome data
FR 40	The system must support mobile-responsive interfaces allowing field technicians to view work orders and equipment details from field locations

Project Functional Requirements

ID	Description
FR 41	The system must maintain a complete audit trail of all user access, alerts generated and operator actions — retained for a minimum of 3 years for regulatory compliance
FR 42	The system must interface with AEP's existing SCADA systems without requiring replacement of operational grid control systems — connecting via standard utility protocols only
FR 43	The system must provide an API allowing authorized external systems including regulatory reporting tools to query grid performance and compliance data
FR 44	The system must export all reports and data in standard formats (PDF, CSV, Excel) compatible with AEP's existing regulatory reporting workflows
FR 45	The system must integrate with AEP's existing enterprise identity management system supporting single sign-on authentication for all user roles
FR 46	The system must generate real-time load forecasting outputs by geographic zone updated every 6 hours under normal conditions and every 30 minutes during active storm events, predicting electricity demand up to 72 hours ahead using historical SCADA data, weather forecasts, and data center demand commitments
FR 47	The system must track reliability performance at the individual circuit level for circuits serving committed data center customers, recording cumulative outage minutes, SLA compliance status, and year-to-date availability against contracted uptime thresholds for each data center account

Project Functional Requirements

ID	Description
FR 48	The system must record each outage event with the following minimum data fields: event ID, start timestamp, end timestamp, duration in minutes, affected circuit ID, data center customer flag, affected customer count, root cause classification, equipment ID, maintenance work order reference, responding operator ID, and a predicted versus unplanned failure flag that identifies whether the ML model flagged the equipment prior to the outage occurring
FR 49	The system must calculate real-time grid optimization recommendations including optimal generation dispatch adjustments and transmission routing based on current load distribution, equipment health scores, and 72-hour demand forecasts — displaying actionable recommendations to grid operators on the Grid Optimization Dashboard within 60 seconds of significant load changes exceeding 5% of zone capacity
FR 50	The system must attach a confidence score to every ML model prediction. Predictions with confidence scores below 80% must be flagged for human review and must not trigger automated work order generation or operational recommendations without supervisor acknowledgment
FR 51	The system must support pre-synchronization of assigned work orders, equipment health records, and maintenance procedures to field technician mobile devices before field deployment, enabling read access to assigned work items and the ability to record work completion status without active network connectivity
FR 52	The system must enforce write-once protection on confirmed outage event records. Any correction to a confirmed outage record must be recorded as a formal amendment entry referencing the original record, with the correction reason, timestamp, and authorizing supervisor — preserving the original record for regulatory audit purposes

Project NON-Functional Requirements

ID	Category	Description
NFR 01	Performance	The system must process and display real-time sensor data within 60 seconds of capture from SCADA systems
NFR 02	Performance	The system must generate equipment failure alerts within 90 seconds of anomaly detection
NFR 03	Performance	All dashboard pages must load within 3 seconds under normal operating conditions for all user roles
NFR 04	Performance	The system must sustain processing of a minimum 10,000 concurrent sensor data points per second during normal operations
NFR 05	Performance	The system must calculate and publish optimal generation dispatch recommendations and transmission routing adjustments in real time based on current load distribution, equipment health status, and forecasted demand updating grid optimization outputs within 60 seconds of significant load changes across the T&D network.
NFR 06	Performance	The system must handle peak data volumes of 3x normal load during storm events without degradation in alert generation speed
NFR 07	Performance	The system must provide real-time data connections for grid operator dashboards with maximum display latency of 60 seconds from sensor capture to dashboard visualization — achieved through direct database query mode rather than cached data refresh for the Grid Operations and Grid Optimization dashboards

Project NON-Functional Requirements

ID	Category	Description
NFR 08	Throughput	The system must process and store a minimum of 500GB of sensor and operational data per day across the full T&D network
NFR 09	Availability	The system must maintain 99.9% uptime — equivalent to less than 9 hours downtime per year — given its role in critical grid operations
NFR 10	Availability	Field technician mobile interfaces must remain functional for a minimum of 8 hours without network connectivity using pre-synchronized work order and equipment data, with automatic synchronization of recorded work completion data when connectivity is restored
NFR 11	Reliability	The system must include automated failover to backup infrastructure within 60 seconds of primary system failure
NFR 12	Reliability	The system must continue generating alerts from cached data during connectivity interruptions of up to 5 minutes without data loss
NFR 13	Reliability	Predictive models must achieve a minimum 80% precision rate on equipment failure predictions before deployment to production

Project NON-Functional Requirements

ID	Category	Description
NFR 14	Reliability	The system must maintain data integrity with zero tolerance for corruption- all sensor readings must be validated before storage
NFR 15	Reliability	All retrained predictive models must pass a minimum validation threshold of 80% precision on a held-out test dataset before deployment to production. Failed retraining cycles must retain the previous production model and generate an alert to the data science team for manual investigation
NFR 16	Security	The system must comply with NERC CIP security standards for all components connected to operational grid systems
NFR 17	Security	All data transmitted between SCADA systems and the analytics platform must be encrypted using AES-256 encryption
NFR 18	Security	The system must implement role-based access control with multi-factor authentication for all user accounts
NFR 19	Security	The system must log all user access, data queries and configuration changes maintaining a complete audit trail for a minimum of 3 years
NFR 20	Security	The system must undergo quarterly penetration testing and annual NERC CIP compliance audits
NFR 21	Security	The system must isolate operational grid control systems from analytics systems using network segmentation thus preventing analytics platform compromise from affecting live grid operations
NFR 22	Scalability	The system must scale horizontally to accommodate 50% growth in connected sensors without architecture redesign

Project NON-Functional Requirements

ID	Category	Description
NFR 23	Scalability	The system must support expansion to additional AEP operating companies and grid regions without requiring core platform rebuilding
NFR 24	Scalability	Data storage architecture must support 10-year data retention growth without performance degradation
NFR 25	Usability	Grid operator dashboards must be operable by trained staff within 2 hours of onboarding
NFR 26	Usability	The system must support mobile-responsive interfaces for field technicians accessing work orders and equipment data from field locations
NFR 27	Usability	Alert notifications must clearly distinguish severity levels through color coding, iconography and plain language descriptions
NFR 28	Usability	Executive dashboards must present KPIs in non-technical language accessible to C-suite users without operational grid expertise

Project NON-Functional Requirements

ID	Category	Description
NFR 29	Maintainability	Predictive models must be retrained automatically on a monthly basis using new sensor and maintenance outcome data
NFR 30	Maintainability	The system must support configuration of alert thresholds, model parameters and dashboard layouts by authorized administrators without requiring code changes
NFR 31	Maintainability	System updates and patches must be deployable without full system downtime- supporting rolling updates to maintain 99.9% availability
NFR 32	Compatibility	The system must integrate with AEP's existing SCADA systems using standard utility protocols without requiring replacement of operational grid control systems
NFR 33	Compatibility	The system must export reports in standard formats (PDF, CSV, Excel) compatible with AEP's existing regulatory reporting workflows
NFR 34	Compatibility	The system must be compatible with AEP's existing enterprise identity management system for single sign-on authentication
NFR 35	Compatibility	Outage event records must be retained for a minimum of 6 years to comply with NERC reliability reporting requirements and state utility commission audit requirements across all 11 operating states, with records remaining queryable and exportable throughout the retention period

Project: Transmission and Distribution Grid Optimization and Predictive Operations Analytics Platform



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DATA COLLECTION

Apache Kafka

Role: Real-time streaming ingestion

What it collects: SCADA system measurements — voltage, frequency, line flows, breaker status PMU sensor readings — phasors, frequency, power Equipment health alerts — temperature spikes, vibration anomalies Grid alarms — threshold breaches, equipment status changes

Why Kafka for T&D:

- Handles 10,000+ concurrent sensor data points per second
- Absorbs 3x peak data volume during storm events without degradation — no data loss
- If Spark disconnects during a storm, Kafka retains all messages and replays them when connectivity restores
- Horizontal scaling adds brokers as AEP's sensor network grows across II states

Sensitivity: SCADA data reveals real-time grid vulnerability. Topic-level ACLs ensure only authorized Spark consumers can read operational grid data — unauthorized access blocked before data reaches storage.

Apache NiFi

Role: Batch and scheduled ingestion

What it collects:

- Weather forecast data every 6 hours — feeds both load forecasting AND vegetation scoring models
- Historical maintenance records, equipment inspection reports, work order history
- Satellite imagery for vegetation threat analysis along transmission corridors
- Equipment metadata — asset registry updates, sensor calibration records

Why NiFi for T&D:

- Single platform handles all batch source types — REST APIs, file transfers, database queries
- Data quality validation flags sensor readings outside physically possible ranges before storage
- Full data provenance — every record tracked through every step, quarantined if it fails validation
- LDAP integration with AEP's enterprise identity management

Sensitivity: Satellite imagery reveals AEP's transmission corridor layout and infrastructure vulnerability. TLS encryption and role-based access control restrict satellite imagery processors to authorized data engineers only.

Together Kafka and NiFi ensure no operational data enters the platform without validation, encryption, and a complete audit trail.



FRAMEWORK

Apache Spark — Core Processing Engine

Unified engine handling real-time streaming, batch processing, ETL transformation, and machine learning — eliminating the need for separate tools for each workload.

Streaming: Processes real-time SCADA sensor data from Kafka, calculates equipment health scores every 30 seconds, and generates failure alerts within 90 seconds of anomaly detection. During storm events automatically switches outage risk recalculation from 6-hour to 30-minute cadence.

Machine Learning — five predictive models:

- Equipment failure prediction — classifies risk as Critical (24hrs), High (7 days), Medium (30 days), Low (beyond 30 days)
- Load forecasting — predicts electricity demand by geographic zone up to 72 hours ahead using SCADA data, weather forecasts, and data center demand commitments
- Grid optimization — calculates real-time generation dispatch recommendations and transmission routing adjustments
- Vegetation threat scoring — combines TensorFlow satellite imagery outputs with weather and outage history
- SAIDI/SAIFI calculation — aggregates outage event data by region for regulatory reporting

SQL and ETL: Interactive queries for planning engineers on network bottleneck analysis and capital investment prioritization. ETL normalizes sensor readings from seven operating companies' SCADA vendors into consistent formats before storage.

Connectivity: Native integration with Kafka, Kudu, HDFS, Oozie, YARN, and Power BI — no additional connectors required for any core platform tool.

Why Spark: In-memory processing is 9x faster than MapReduce alternatives. YARN integration means storm event resource scaling is fully automatic — no manual intervention when sensor data volume triples

Apache Oozie — Workflow Orchestration

Sequences and schedules all platform jobs automatically — ensuring every pipeline step runs in the correct order with dependencies met before downstream jobs trigger.

Key workflows:

- Daily 6am operational reports — NiFi batch ingestion completes, Spark processes overnight sensor data, Cognos generates reports, delivered to operations managers by 6am.
- Monthly model retraining — validates new training data in HDFS before triggering Spark MLib retraining. Models must pass 80% precision threshold before replacing production versions. Failed retraining retains previous model and alerts the data science team.
- Storm event activation — NiFi detects severe weather alert, Oozie triggers storm preparation mode, Spark switches to 30-minute recalculation cadence, YARN reallocates resources from lower-priority queues to grid monitoring.
- Regulatory reporting — monthly SAIDI/SAIFI data compiled from Kudu Outage domain, Cognos generates formatted reports for all 11 state utility commissions, distributed to regulatory affairs team.

Why Oozie: Without orchestration every pipeline step requires manual triggering — unsustainable for a 24/7 critical infrastructure platform. Workflow definitions stored in HDFS with version control and audit logging. Any unauthorized workflow change requires HDFS write permissions and is permanently logged

TensorFlow — Vegetation Imagery Analysis

Selected specifically because vegetation threat scoring is an image recognition problem that tabular ML tools including Spark MLlib cannot solve. Satellite imagery arrives as unstructured multispectral image files — TensorFlow is the only tool in the architecture capable of extracting structured threat scores from this format.

- **Why this matters for AEP:** Vegetation is the cause of approximately 25% of transmission outages nationally. AEP's subsidiary SWEPCO already uses aerial imagery for vegetation management. TensorFlow extends this to satellite scale — covering 40,000+ miles cost-effectively without helicopter inspections. Industry deployments achieve up to 90% accuracy identifying trees presenting strike risk to power lines.
- **Monthly retraining:** New satellite imagery and field crew validation data — confirmed trimming outcomes feed back into the training dataset improving accuracy over time.
- **Security:** All TensorFlow model files are cryptographically signed. Before any model is loaded for production inference its signature is verified — preventing model poisoning attacks where corrupted models produce false vegetation threat outputs.
- **Connectivity:** NiFi delivers satellite imagery to HDFS. TensorFlow reads imagery for training and inference. Outputs structured threat scores to Spark for combination with weather and outage history. Oozie schedules monthly retraining. YARN allocates GPU compute resources for training jobs, isolating them from CPU-based Spark workloads



STORAGE

Apache Kudu- Active Sensor Data Storage

Why Kudu for T&D: Grid operators need millisecond random access to live equipment readings. Data scientists need fast analytical scans across 18 months of sensor history for ML training. Kudu delivers both on the same data — no ETL pipeline copying between a fast lookup store and an analytical store.

- **Seven storage domains:** Operations, Sensor, Equipment, Maintenance, Outage, Weather, Vegetation — each domain independently encrypted and access-controlled per user role.

Two critical data structures:

- Data center circuit mapping — Equipment domain tags every circuit serving a committed data center customer with contracted capacity, reliability SLA, and cumulative outage minutes. Directly supports the \$560,000 data center retention ROI stream.
- Outage event schema — each record captures 12 fields including a predicted versus unplanned flag. This flag is the evidence that validates the \$800,000 outage prevention ROI — proving the platform predicted the failure before it occurred.

Scalability: Horizontal scaling — adding tablet servers proportionally increases capacity and throughput. Supports 50% sensor network growth without architecture redesign.

Security: Kerberos authentication. Column-level access control enforcing the role-to-domain matrix. HDFS Transparent Encryption at rest. SSL/TLS in transit.

Connectivity: Native Spark, NiFi, and Power BI integration — no additional connectors required

HDFS- Historical Archives and ML Training Data

Why HDFS for long-term storage: Optimized for large sequential reads — exactly how Spark reads months of sensor history in one pass for ML training. At 500GB of sensor data per day HDFS horizontal scaling absorbs decade-scale growth on commodity hardware.

What HDFS stores:

- Historical SCADA archives for ML model training
- Raw satellite imagery before TensorFlow processing
- Completed maintenance records and inspection reports
- Historical weather data for load forecasting model training
- Regulatory archive — outage records retained 6 years per NERC reliability requirements

Encryption: Each storage domain has its own encryption zone and key — a breach in one zone does not expose other zones' data.

How Kudu and HDFS work together: Active operational data in Kudu — last 18 months. Data older than the active window archives to HDFS automatically through nightly Oozie workflows. ML models train on HDFS historical data and deploy predictions back into Kudu for operational use.

Connectivity: Native Spark, TensorFlow, NiFi, Oozie, and Chukwa integration



BI TOOLS

Power BI

On-premises Report Server — all grid data stays within AEP's network perimeter. NERC CIP compliant. No data leaves the organization.

Two connection modes chosen for different operational needs:

- DirectQuery — grid operators only. Live data, maximum 60-second latency. Always current during storm events when equipment status changes by the minute.
- Import mode — all other users. Scheduled refresh every 5-15 minutes. Fast performance for complex geographic analysis and historical trend queries.

Eight dashboards:

- Grid Operations — real-time voltage, frequency, line flows
- Grid operators. Equipment Health — condition scores, failure timelines
- Maintenance teams. Outage Prediction — at-risk equipment by failure probability
- Operations managers. Preventive Maintenance — condition-based work orders
- Maintenance technicians. Network Optimization — bottleneck heat maps, capital investment prioritization
- Planning engineers. Grid Optimization — dispatch recommendations, transmission routing. Control center
- Data Center Reliability — SLA compliance, outage minutes per account. Account managers. Directly supports \$560K retention ROI stream.
- Executive — outages prevented, savings, ROI actuals in non-technical language. C-suite.

Mobile: Pre-synchronized offline access for field technicians — 8 hours without connectivity.

Security: MFA, row-level security by geographic region, 15-minute session timeout for grid operators.

Cost advantage: Lowest cost enterprise BI tool — viable for 200+ users across 7 operating companies

IBM Cognos- Regulatory and Compliance Reporting

Why Cognos and not Power BI for regulatory reporting: Power BI produces interactive dashboards. Cognos produces pixel-perfect paginated documents — the format state utility commissions and NERC require. Two different outputs for two different audiences.

Why Cognos specifically:

- Framework Manager defines SAIDI, SAIFI, and reliability metrics once — every report across all 11 states calculates them identically. No metric inconsistency across state submissions.
- Cognos Application Firewall validates every request for SQL injection and malicious patterns before execution — application-level security beyond network firewalls.
- Scheduled distribution with automatic retry — no missed regulatory deadlines due to system failures.

Four report types:

- Monthly SAIDI/SAIFI reliability reports — all 11 state utility commissions
- NERC CIP compliance reports — security controls, audit trail summaries
- Quarterly ROI reports — actual benefits realized against projected figures across all four ROI streams
- Daily 6am operational reports — alerts triggered, maintenance actions, outages prevented



SECURITY

HDFS- Historical Archives and ML Training Data

Protecting Critical National Infrastructure

- AEP's T&D platform handles SCADA data revealing real-time grid vulnerabilities — which equipment is stressed, which circuits are near capacity, which transformers are approaching failure. This is not business data. It is critical national infrastructure data regulated under NERC CIP federal standards.

Three threats the architecture defends against:

- Confidentiality — unauthorized read access reveals grid vulnerability patterns to attackers
- Integrity — unauthorized write access corrupts sensor readings, causing ML models to make dangerous recommendations on poisoned data
- Availability — denial of service against the platform removes grid operators' visibility during the most critical moments

Security is applied at every layer:

Layer	Primary Security Tools
Data Collection	Kafka TLS/SASL/ACL, NiFi TLS/LDAP/Provenance
Storage	Kudu/HDFS Kerberos, Column-level ACL, Encryption at rest
Framework	Spark/Oozie Kerberos, TensorFlow model signing
BI	Power BI MFA/RLS, Cognos CAF Firewall/LDAP
Platform	ZooKeeper SASL, YARN Queue ACL, Chukwa Audit Logs

Security Measures Per Tool

Data Collection Layer

- Kafka — TLS encryption on all data in transit between SCADA systems and the analytics platform. SASL authentication requires every producer and consumer to authenticate before connecting. Topic-level ACLs restrict which services read or write which topics — Spark Streaming can read sensor topics but cannot write to them. All authentication events logged to HDFS audit trail retained 3 years.
- NiFi — TLS for all data flows. LDAP integration with AEP's enterprise identity management for single sign-on. Data provenance tracks every record through every processor — when it arrived, what transformation was applied, where it was delivered. Unauthorized pipeline modifications require HDFS write permissions and are permanently logged.

Storage Layer

- Kudu — Kerberos authentication for all access. Column-level access control enforces the role-to-domain matrix — a maintenance technician's credentials only grant access to Equipment and Maintenance columns for their assigned equipment. SSL/TLS in transit. HDFS Transparent Encryption at rest.
- HDFS — Transparent Encryption at rest with separate encryption zones per storage domain. Each domain has its own encryption key — a breach in one zone does not expose other zones. Kerberos authentication. POSIX ACLs and HDFS ACLs for file permissions. NameNode audit logging records every file access with user identity, timestamp, and IP address.

Security Measures Per Tool

Framework Layer

Spark: Kerberos authentication for all job submissions. SSL/TLS between driver and executors. Column-level permissions enforced at query level — a maintenance technician's Spark session cannot return Operations domain data even if the SQL attempts it.

Oozie: workflow definitions stored in HDFS protected by ACLs. Only system administrators can modify workflow definitions. HTTPS for web interface. Every workflow execution logged — who triggered it, what jobs ran, what data was accessed.

TensorFlow: model files cryptographically signed. Signature verified before any model loads for production inference — prevents model poisoning attacks. TLS encrypts all data transfer to and from the serving layer. GPU resources isolated per job.

BI Layer

Power BI : Azure Active Directory with MFA. Row-level security enforces geographic scoping. Session timeouts 15 minutes for grid operators, 30 minutes all other roles. Full audit logging of access, queries, and exports.

Cognos: Application Firewall validates every request for SQL injection and cross-site scripting before execution. LDAP authentication. All report access and generation events logged. Automated reports run under service account credentials with read-only domain access

Security Measures Per Tool

Platform Layer

ZooKeeper: SASL/Kerberos authentication. ACLs restrict which services access which ZooKeeper nodes. SSL/TLS for all client-server communication. If ZooKeeper is compromised every dependent service is at risk

YARN: Kerberos for all job submissions. Queue-level ACLs prevent unauthorized queue access. Container isolation prevents jobs accessing other jobs' memory. All job submissions logged.

Chukwa: collects infrastructure logs from all platform servers. Routes to HDFS for 3-year retention. HICC provides platform health visibility. Supports NERC CIP audit trail requirements

NERC CIP Compliance Strategy

NERC CIP Standard	Requirement	How Addressed
CIP-005 Electronic Security Perimeters	Network segmentation isolating grid control from analytics	Kafka topic ACLs, YARN queue isolation, network segmentation between SCADA and analytics layers
CIP-007 System Security Management	MFA, session management, patch management	Power BI and Cognos MFA, session timeouts, ZooKeeper and YARN security controls
CIP-010 Configuration Change Management	Document and control all configuration changes	Oozie workflow version control in HDFS, NiFi flow change audit trail, ZooKeeper configuration registry
CIP-011 Information Protection	Protect sensitive bulk electric system data	AES-256 encryption in transit across all tools, HDFS encryption zones at rest, column-level access control in Kudu and Spark
CIP-013 Supply Chain Risk Management	Manage security risks in vendor relationships	TensorFlow model signing, open-source Apache Foundation tools with active security patch programs, IBM and Microsoft enterprise support contracts with defined security SLA

Access Control Matrix

Domain	Grid Operator	Maint. Tech	Planning Eng	Sys Admin	Executive	Vegetation Crew	SCADA Eng
Operations	Full	None	Historical	None	Aggregated	None	None
Sensor	Full	Assigned	Historical	None	None	None	Read only
Equipment	Full	Assigned	Full	None	None	Assigned lines	None
Maintenance	Read only	Assigned	Full	None	Aggregated	None	None
Outage	Full	None	Full	None	Aggregated	None	None
Weather	Full	None	Full	None	None	None	None
Vegetation	Alerts only	Work orders	Full	None	None	Full assigned	None
Audit logs	None	None	None	Full	None	None	None



MANAGEABILITY

Zookeeper

Platform Coordination and Cluster Health

ZooKeeper is the coordination backbone — every major platform tool depends on it for health awareness, leader election, and automatic failover.

What it coordinates for AEP's architecture:

- Kafka broker failover — partition leader re-election when a broker fails. Sensor data continues flowing without human intervention.
- HDFS NameNode high availability — automatic failover to standby within 60 seconds
- Kudu Master high availability — master leader election

Live service registry: Real-time awareness of which services are alive and where they are running across the entire cluster.

Four-letter health commands: Built-in diagnostics polled automatically for proactive alerting — active connections, outstanding requests, latency statistics.

Deployed first: ZooKeeper must be secured before any dependent tool. A compromised ZooKeeper compromises every tool that depends on it.

YARN

Three priority queues:

- Grid Operations — guaranteed resources, cannot be preempted. Real-time Spark Streaming, Power BI DirectQuery, storm event jobs.
- Analytics — elastic resources. Planning engineer SQL queries, batch processing, Cognos report generation.
- Model Training — low priority, preemptible. Monthly MLlib retraining, TensorFlow vegetation model training.

Storm event scaling: Oozie triggers storm preparation mode — YARN automatically reallocates resources from Model Training to Grid Operations. Spark gets additional executors to handle 3x peak sensor volume. No manual intervention required.

Node health monitoring: If a node's disk or CPU reaches critical levels YARN marks it unhealthy and stops scheduling containers on it — preventing cascading failures.

Application lifecycle: If a Spark Streaming job crashes YARN attempts automatic restart before alerting administrators

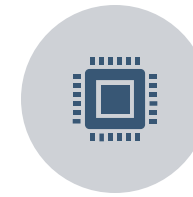
Chukwa



Chukwa collects logs, metrics, and health signals from every platform server — surfacing them through HICC, the Hadoop Infrastructure Care Center dashboard.



How it works: Agents on every server collect system logs, application errors, and performance metrics continuously. Collectors write to HDFS. ETL processes structure the raw data. HICC displays health trends, error rates, and infrastructure alerts to platform administrators.



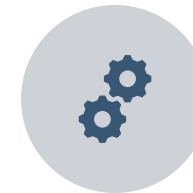
What administrators see: Server health — CPU, memory, disk, network across all nodes
Application errors — Kafka consumer lag, Spark failures, NiFi processor errors
Performance trends — gradual degradation identified before failure occurs



NERC CIP audit support: All platform server logs routed to HDFS, retained 3 years minimum — infrastructure audit trail for compliance audits.



Each tool's native real-time interface: Kafka UI — consumer lag, broker health
Spark UI — streaming latency, job duration
YARN console — queue utilization, node health
NiFi canvas — processor health, pipeline status
Kudu Master UI — tablet distribution, write latency



Chukwa aggregates the historical picture. Each tool's native interface handles real-time monitoring. AEP's enterprise monitoring infrastructure handles cross-tool metric aggregation.

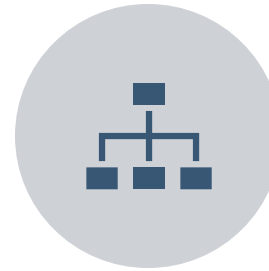
Additional Tools



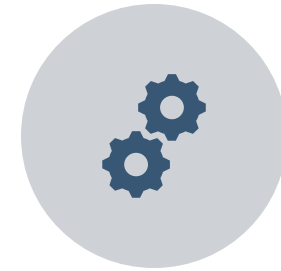
ZOOKEEPER: GIVES KAFKA, HDFS, AND KUDU AUTOMATIC FAILOVER THEY COULD NOT ACHIEVE INDEPENDENTLY. PLATFORM-WIDE RESILIENCE FROM A SINGLE COORDINATION LAYER



YARN: GIVES SPARK, TENSORFLOW, AND OOZIE GUARANTEED RESOURCES AND STORM-EVENT SCALING WITHOUT CUSTOM ENGINEERING. OPERATIONAL PRIORITIES ENFORCED AUTOMATICALLY



CHUKWA: GIVES THE ENTIRE PLATFORM A UNIFIED NERC CIP-COMPLIANT AUDIT TRAIL NO SINGLE TOOL COULD PROVIDE ALONE. INFRASTRUCTURE MANAGEMENT BECOMES PROACTIVE, NOT REACTIVE



TOGETHER THEY ARE WHAT MAKES THIS A PRODUCTION-GRADE ENTERPRISE PLATFORM — NOT A COLLECTION OF INDIVIDUAL TOOLS